

# Karthik Pasupuleti

✉ karthikpasupuleti@gmail.com | 📞 +91 7601024711 | 🔗 LinkedIn | 🐙 GitHub | 🌐 Portfolio

## Professional Summary

Analytical and impact-driven **AI/ML engineer** skilled in building **reliable, scalable solutions** across data automation, software development, and intelligent systems. Recognized for delivering effective **Software and ML** implementations, supported by **strong communication, impactful research, and national-level hackathon wins** demonstrating **leadership and innovation**.

## Education

### NRI Institute of Technology

*B.Tech in Artificial Intelligence and Machine Learning*

Aug 2023 – June 2026

**CGPA: 8.8/10.0**

### Dhanekula Institute of Engineering and Technology

*Diploma in Electronics and Communication Engineering*

June 2020 – June 2023

**CGPA: 9.2/10.0**

## Technical Skills

**Programming & Web Dev:** Python (PyTorch, TensorFlow, Keras, Scikit-learn, Flask, Streamlit), HTML, CSS, JavaScript (Basic)

**Machine Learning & AI:** Transformers (LLM's, GPT), Generative AI, Natural Language Processing, Computer Vision,

Convolutional Neural Networks, Retrieval-Augmented Generation, Transfer Learning, Quantum Machine Learning (Basic)

**Data Analytics & Visualization:** Data Cleaning, EDA, Structured Query Language, Statistics (Basic), Pandas, Matplotlib, Seaborn

**Platforms & Tools:** GitHub, Hugging Face, IBM Quantum Platform, Colab, MS Excel, PowerBI, DBeaver, VS Code, Anaconda

**Database & Deployment:** MySQL, MongoDB, PostgreSQL, Docker, Render, Netlify, Vercel

## Professional Experience

### Centre for Development of Advanced Computing (C-DAC)

*AI & ML Intern*

Hyderabad, India

Aug 2025 – Feb 2026

- Conducted **research, debugging, and hyperparameter tuning** on **Hybrid Quantum-Classical anomaly classification models**, achieving **88–97%** accuracy compared to **73–96%** from **classical CNN** baselines across benchmark datasets.
- Orchestrated complete design and benchmarking of **Quantum-Enhanced CNN research**, delivering validated results which reduced model training time by **15%** and improved accuracy by **4%** in **anomaly classification tasks**.

### AP State Civil Supplies Corporation Limited (APSCSC)

*Project Intern*

Vijayawada, India

June 2025 – Aug 2025

- Led automation for **large-scale English to Telugu data translation (20+ lakh rows)** for **official govt. records** preparation, achieving **85%** automation efficiency and collaborating with **Government officials** to support **data-backed decisions**.
- Implemented 3 lightweight web applications such as **MLS Point Locator**, **Route Mapping System**, **FPS Proximity Analyzer** (Geopy + OSRM), and 1 model, **Telugu Auto-Correction models** to enhance localization workflows.

### Efftronics Systems Private Limited

*System Administrator Intern*

Mangalagiri, India

Dec 2022 – June 2023

- Managed company-wide **IT support** and enhanced **infrastructure** across multiple sites by resolving **hardware/software issues, mapping network architecture**, and implementing automated monitoring, reducing response times by **20%**.
- Facilitated system operations through **OS installations** and contributed to **cloud-integrated data workflow development**, handling **data transfer** from external sources to internal servers to complete roughly **30% of the implementation**.

## Projects

### AutoGrader: AI-Powered Answer Evaluation System

🔗 Code Link

*Tech Stack: Python, NLTK, PyMuPDF, SBERT, Cross-Encoder, Gemini OCR, MongoDB, Flask*

- Built **AutoGrader**, an AI-powered system for handwritten exam evaluation using **SBERT and Cross-Encoder**, achieving **weighted ensemble scoring (0.4/0.6)** for nuanced semantic assessment.
- Delivered **large-scale answer evaluation** by engineering an end-to-end pipeline with **Google Gemini, NLTK, and Flask**.

### Quantum-Enhanced CNN for Image Anomaly Classification

🔗 Code Link

*Tech Stack: Python, PyTorch, TorchVision, PennyLane-Lightning, Custom CNN, VQC*

- Developed a **Hybrid Quantum-Classical CNN** that incorporates a six-qubit **Variational Quantum Circuit** as the classifier module, enhancing decision boundaries and achieving a **3–6% accuracy lift** over **classical CNN baselines**.
- Achieved **95.73% accuracy (Aluminium Foil)**, **97.45% (Chest X-Ray)**, and robust results on **BUSI (78.85%)** and **MVTec Mesh (88.57%)**, outperforming **classical CNNs**, especially in **low-data regimes**.

## Publications & Achievements

- Patents:** AI-IoT Dual-Stage Crowd Risk Detection; AI for Adolescent Health Monitoring, *The Patent Office Journal (India)*.
- Publication:** Automated Monitoring and Real-Time Alert System for DWLRs; Short-to-Long: An AI Powered Pipeline for Video-to-Reel Summarization; AutoGrader: ML & NLP-Based Handwritten Exam Evaluation System, *IEEE*.
- Hackathon:** Winner in Four National-level 24-hour Hackathons, *Innovators Fusion; Prajwalan; Teckzite; Research Conclave*.

## Certifications

Machine Learning Specialization, **DeepLearning.AI** | CS50's Introduction to Computer Science, **HarvardX** | Data Analytics Specialization, **Google** | Generative AI Fundamentals, **Databricks** | IT Automation with Python Specialization, **Google**